Assignment 1: Overview of Visualizations

Principles of Data Visualization

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1 Visualization 1 - Grouped Bar Chart

Image:

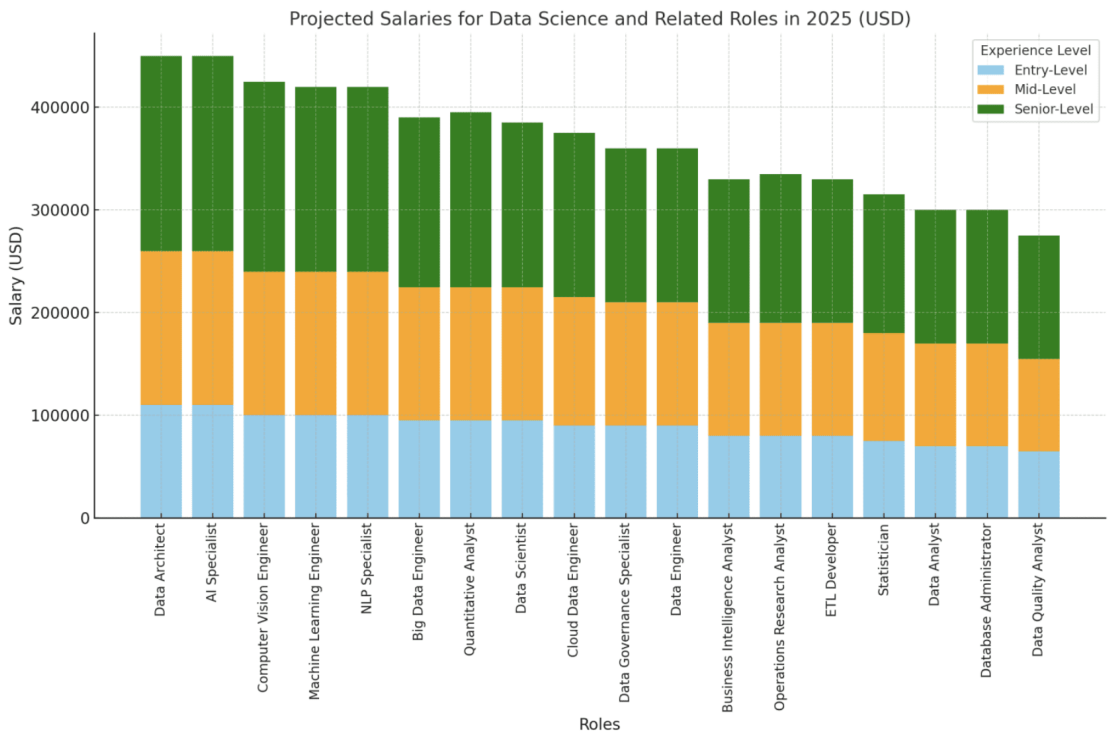

Figure 1: Data Science Salaries by Role

Source: https://www.kdnuggets.com/data-science-salaries-job-market-analysis-2024-2025

Chart type: Horizontal Grouped Bar Chart (salary ranges by job role and experience level)

1.1 Mapping Analysis

What's being shown	Data type	Channel used
Job roles (Data Scientist, ML Engineer, Data Analyst, etc.)	Nominal	Position (Y-axis)
Experience levels (Entry, Mid, Senior)	Ordinal	Color hue, grouped bars
Salary amount (USD)	Quantitative	Bar length (position on X-axis)
Salary labels	Quantitative	Text annotations on bars

1.2 Effectiveness Note

The grouped horizontal bars allow for clear comparison between experience levels within each role. The color differentiation for levels is effective, but the overall readability could improve with consistent ordering of levels and spacing between role groups to avoid visual crowding.

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2 Visualization 2 - Multi-line Chart

Image:

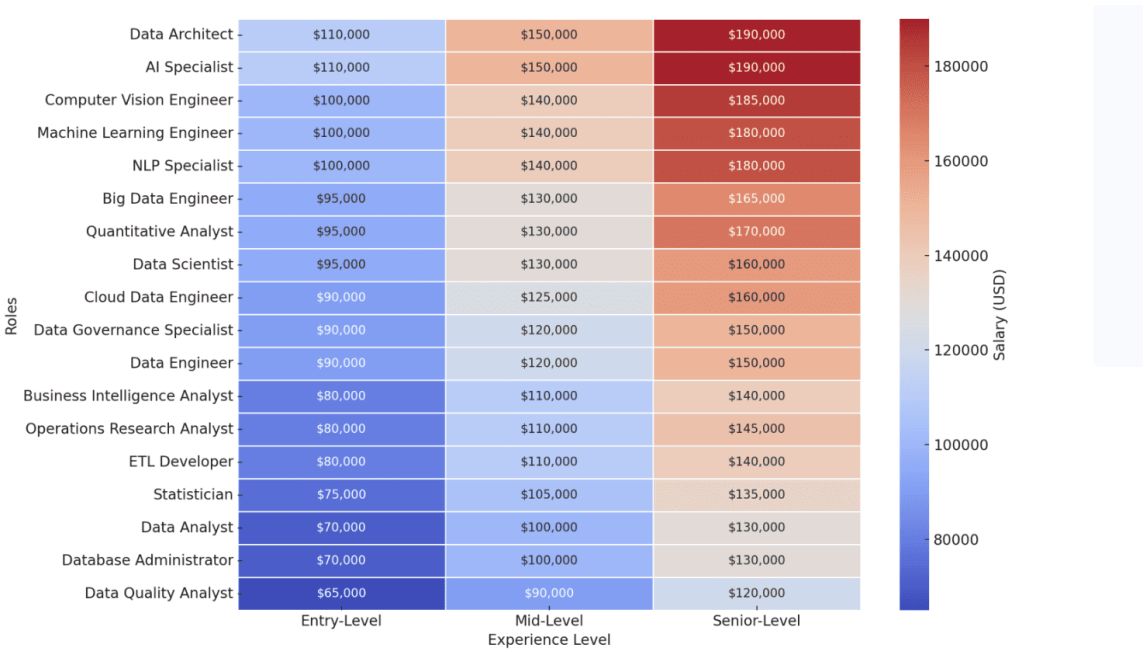

Figure 2: Salary Trends by Experience Level

Source: https://www.kdnuggets.com/data-science-salaries-job-market-analysis-2024-2025

Chart type: Multi-line Chart (Average salary over time)

2.1 Mapping Analysis

What's being shown	Data type	Channel used
Year (2020–2024)	Temporal	Position (X-axis)
Average salary (USD)	Quantitative	Position (Y-axis), line height
Experience level	Ordinal	Color hue, line style
Salary trend	Quantitative	Line slope/angle

2.2 Effectiveness Note

The separate color-coded lines clearly display progression across experience levels. The viewer easily sees that senior and executive salaries increase faster than entry-level ones. Adding data labels or smoothing could improve the precision of interpretation without clutter.

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3 Visualization 3 - Heatmap

Image:

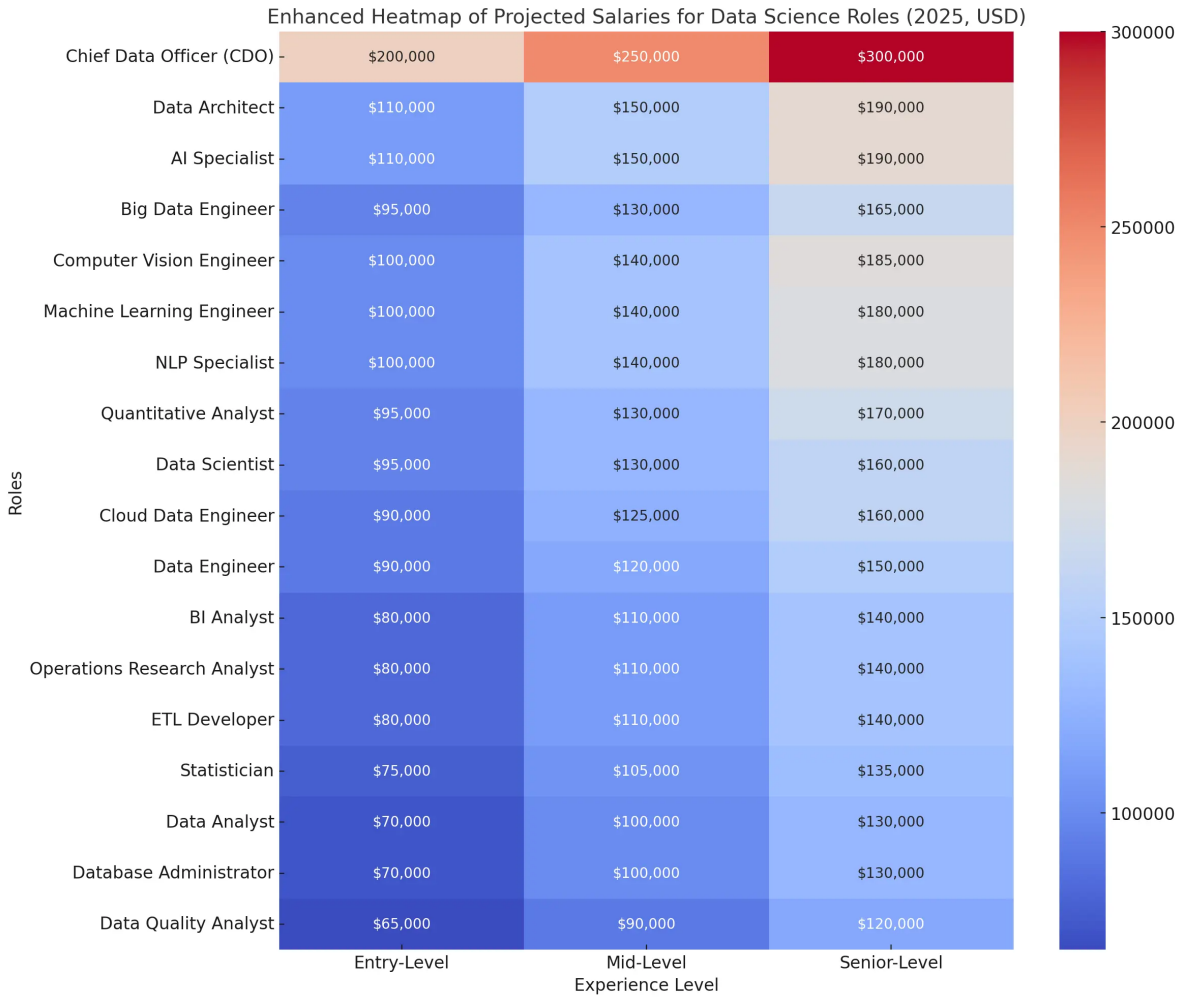

Figure 3: Salary Heatmap by Role and Experience

Source: https://www.refontelearning.com/salary-guide/data-science-salaries-unveiled

Chart type: Heatmap (Matrix showing salaries by role and experience)

3.1 Mapping Analysis

What's being shown	Data type	Channel used
Job roles	Nominal	Position (Y-axis)
Experience levels	Ordinal	Position (X-axis)
Salary (USD)	Quantitative	Color intensity (dark = higher salary)

What's being shown	Data type	Channel used
Numeric value	Quantitative	Text label (optional)

3.2 Effectiveness Note

The sequential color scale effectively highlights the salary gradient across roles and experience levels. Darker tones draw attention to top-paying roles, such as Data Architect and Chief Data Officer. A color legend or numeric scale would further improve interpretability.

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4 Visualization 4 - Stacked Bar Chart

Image:

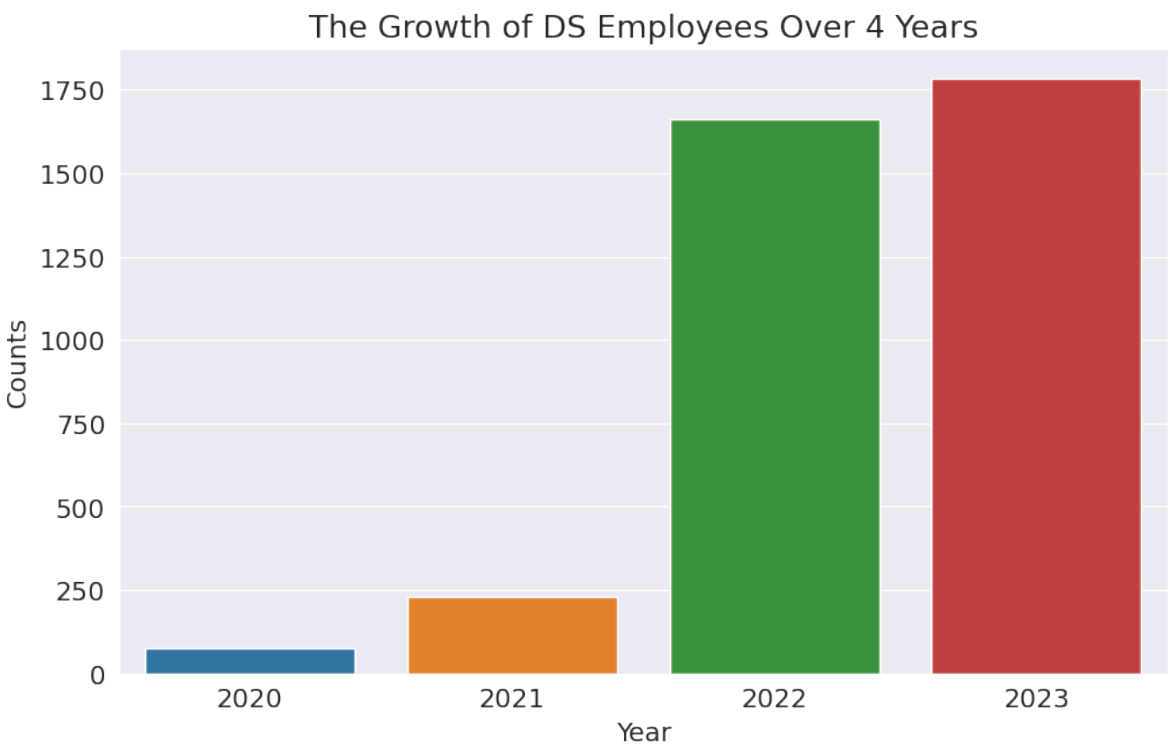

Figure 4: Distribution of Data Science Jobs

Source: https://www.kaggle.com/code/mohamedmagdy191/data-science-salaries-eda-visualization

Chart type: Horizontal Stacked Bar Chart (100% normalized)

4.1 Mapping Analysis

What's being shown	Data type	Channel used
Job titles (8–10 roles)	Nominal	Position (Y-axis)
Experience level (EN, MI, SE, EX)	Ordinal	Bar segments, color hue
Share of experience levels	Quantitative	Segment length (width)
Total composition	Quantitative	Full bar = 100%
Years 2020-2023	Temporal	Position (x-axis)

4.2 Effectiveness Note

The 100% stacked format shows workforce composition across roles effectively. Roles such as Data Scientist have balanced experience distribution, while ML Research Scientist is dominated by senior/executive positions. Color-coding is intuitive, though adding labels would make small segments easier to interpret.

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5 Visualization 5 - Pie Chart

Image:

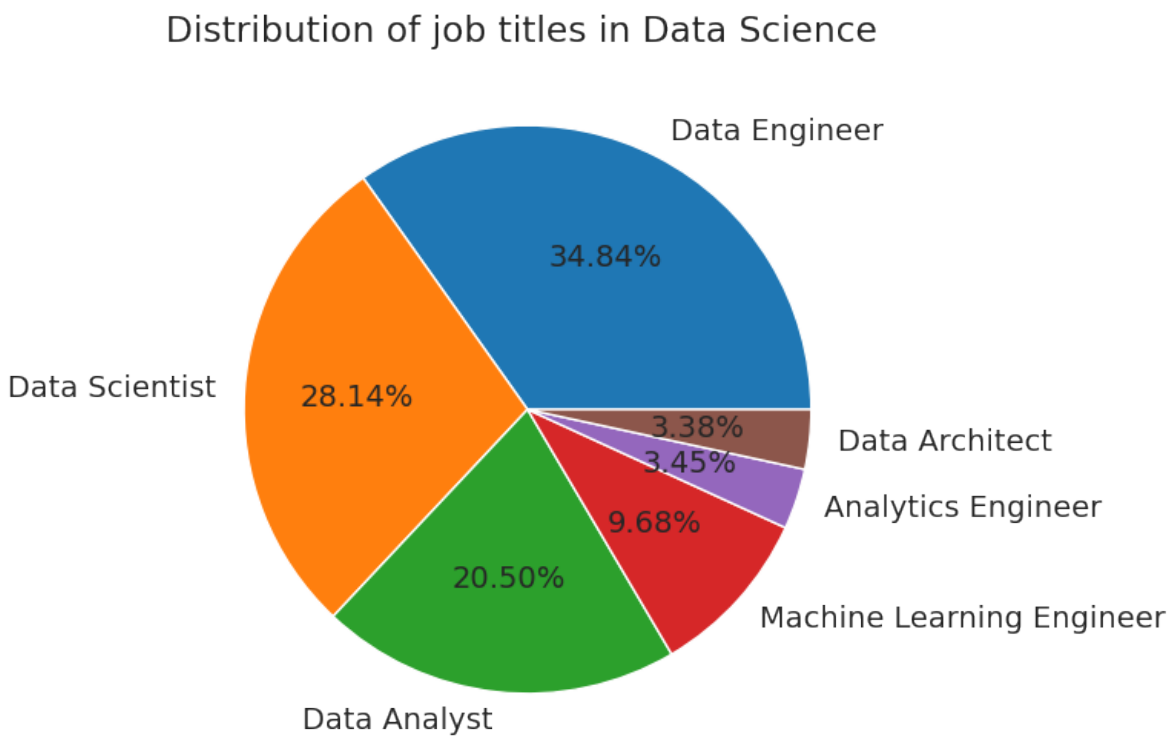

Figure 5: Distribution of Job Titles in Data Science

Source: https://www.kaggle.com/code/mohamedmagdy191/data-science-salaries-eda-visualization

Chart type: Pie Chart with percentage labels

5.1 Mapping Analysis

What's being shown	Data type	Channel used
Job titles (Data Engineer, Data Scientist, etc.)	Nominal	Slice/sector position
Percentage of total jobs	Quantitative	Slice angle and area total of 100%
Percentage labels	Quantitative	Text values at the middle of the sector
Categories	Nominal	Color hue
Categories	Categorical	From largest percentage to least in non clockwise order

5.2 Effectiveness Note

The pie chart immediately communicates that Data Engineer and Data Scientist dominate the market, comprising roughly two-thirds of all roles. The descending slice sizes are easy to read. Still, small slices are difficult to compare — a sorted horizontal bar chart could provide a more accurate visual ranking.